RAJALAKSHMI ENGINEERING COLLEGE
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CS23431 - OPERATING SYSTEM

EXP 7: LRU Page Replacement Algorithm Using C

QUESTION :

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

CODE:

```
#include <stdio.h>

int main() {
   int n, frames;

   printf("Enter number of pages in reference string: \n");
   scanf("%d", &n);

   int ref[n];
   printf("Enter the reference string: \n");
   for(int i = 0; i < n; i++) {
       scanf("%d", &ref[i]);
   }

   printf("Enter number of frames: \n");
   scanf("%d", &frames);

   int frame[frames], time[frames];
   int faults = 0, hits = 0, t = 0;

   // Initialize frames
   for(int i = 0; i < frames; i++) {
       frame[i] = -1;
       time[i] = 0;
   }

   for(int i = 0; i < n; i++) {
       int found = 0;

       // Check HIT
       for(int j = 0; j < frames; j++) {
           if(frame[j] == ref[i]) {
```

```
                hits++;
                t++;
                time[j] = t; // update recent use
                found = 1;
                break;
            }
        }

        // FAULT
        if(!found) {
            int min = 0;

            // Find least recently used
            for(int j = 1; j < frames; j++) {
                if(time[j] < time[min])
                    min = j;
            }

            frame[min] = ref[i];
            t++;
            time[min] = t;
            faults++;
        }
    }

    printf("\nTotal Page Faults: %d\n", faults);
    printf("Total Page Hits: %d", hits);

    return 0;
}
```

Output:

Custom Test Case

Custom Test — Success

Input :

```
5
1 2 3 4 5
5
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```